# Anamta Karim

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## Objective

I enjoy building AI solutions to real-world problems and understanding what makes them work beyond a successful demo. Through projects in NLP, computer vision, and generative AI, I've become increasingly interested in the engineering behind AI systems, from how they process information to where they fall short. I'm drawn to the intersection of AI/ML, data science, and analytics, where technical problem-solving and working with data come together to build something genuinely useful.

## Education

**Jamia Hamdard University** — New Delhi, India
*B.Tech. in Computer Science and Engineering (Artificial Intelligence)* — *2023 – 2027 (Expected)*
- CGPA: 9.25/10

## Projects

**SkinWise – Live Skincare Ingredient & Price Intelligence Platform** — github.com/anamta-karim/SkinWise
Python, Flask, Sentence-Transformers, EasyOCR, Groq LLM, SerpApi

- Built a full-stack AI-powered skincare platform with four modules: Ingredient Analyzer, Personal Regimen, EthiScan, and Ingredient Pulse, combining ingredient safety analysis, personalised routines, ethical brand verification, and search-trend analytics.
- Rebuilt the data layer using four SerpApi engines (Google Shopping, Search, News, and Trends), replacing a curated database of 105 products with live product listings and prices; implemented disk caching, retries, and fallback handling to reduce repeated API calls.
- Engineered a multi-stage ingredient retrieval pipeline with brand and product-variant validation, candidate-list detection, and database checks; evaluated 12 products, retrieving usable ingredient lists for 8, including 7 from exact product pages and 1 from a closest variant.
- Implemented semantic matching using Sentence-Transformers embeddings and cosine similarity to map OCR and web-extracted ingredient names across 966 indexed terms to a 385-ingredient safety database; integrated EasyOCR and Groq-hosted LLMs for label analysis and personalised recommendations.
- Added confidence-aware reporting to flag incomplete ingredient lists and avoid unsupported safety conclusions; wrote 27 pytest unit tests with mocked API responses and containerised the application with Docker.

**Retinal Image Clinical Report Generation using LoRA-Adapted Vision-Language Models** — In Progress
PyTorch, Hugging Face Transformers, PEFT/LoRA, bitsandbytes, Flask, Docker

- Developing a vision-language pipeline to generate free-text clinical descriptions from retinal fundus images using parameter-efficient fine-tuning of pretrained BLIP/InstructBLIP-family models, extending the DeepOpht retinal report-generation research (Huang et al., WACV 2021).
- Configured a reproducible CUDA-enabled training environment and modular pipeline for data handling, model training, and evaluation; designed the setup around a 4 GB VRAM constraint using 4-bit quantization and LoRA adapters to reduce trainable parameters.
- Working with the licensed DeepEyeNet dataset of approximately 15,700 retinal images paired with ophthalmologist-written descriptions; planning BLEU, ROUGE, and CIDEr evaluation against the published DeepOpht baseline, with Flask and Docker deployment planned.

## Technical Skills

**Programming:** Python, SQL
**Data Analysis & Visualization:** Excel, Power BI, Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Cleaning
**Machine Learning:** Scikit-learn, Regression, Classification, Clustering, Data Preprocessing, Feature Scaling, Model Evaluation, Cross-Validation
**AI, NLP & GenAI:** Sentence Transformers, Semantic Embeddings, EasyOCR, LLM Integration, Retrieval-Augmented Generation (RAG)
**Databases & Data Collection:** MySQL, REST APIs, Web Scraping (Requests, Beautiful Soup)
**Development Tools:** Flask, Git, GitHub, Jupyter Notebook, VS Code

## Experience & Leadership

**Data Analytics Job Simulation** *May – June 2026*
Deloitte Australia (via Forage)

- Analyzed business datasets in a forensic technology job simulation to identify patterns and derive actionable business insights.
- Used Excel to classify data and draw business conclusions, then developed a Tableau dashboard to communicate findings and support data-driven decision-making.

**Vice-President** *March – October 2025*
Ignite Community, Jamia Hamdard

- Led a student community of 100+ members, overseeing day-to-day operations, coordinating organizational activities, and supporting collaboration across the team.
- Organized community events and led engagement initiatives, managing planning and execution to encourage participation and strengthen involvement.

**Joint Secretary – Ruling** *2025*
Aarambh'25 – National Student Parliament, Jamia Hamdard

- Coordinated 100+ ruling-side delegates at a national-level student parliament organized by Jamia Hamdard and Unnat Bharat Abhiyan, managing participant communication and coordination throughout the event.
- Managed official documentation and written communications, while overseeing delegate logistics and coordinating event execution with the organizing team.

## Certifications and Courses

**AI/ML for Geodata Analysis** *2026*
*Indian Institute of Remote Sensing (IIRS), ISRO*

**Agentic AI on Google Cloud** *2026*
*Google Cloud Skills Boost – Audited*

**The Ultimate Job Ready Data Science Course** *2026*
*CodeWithHarry – Audited*

**McKinsey Forward Program** *2026 – In Progress*
*McKinsey & Company*

## Achievements

**Reliance Foundation Undergraduate Scholarship** *2023 – Present*
*Reliance Foundation*

- Merit-based undergraduate scholarship recognizing academic performance.

**Smart India Hackathon 2025**

- Cleared the college-level selection round.

**National Space Day 2024** *August 2024*
*Department of Space, Government of India*

- Invited to participate in the National Space Day celebrations commemorating Chandrayaan-3's success, held at Bharat Mandapam, New Delhi.

**Career Acceleration Bootcamp** *August 2024*
*LedBy Foundation, in partnership with the U.S. Embassy, New Delhi*